

WASP-39b

R-1826

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_230604 · created 2026-08-03T05:20:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460099.8233 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.055
Transit depth (Rp/Rs)^2	1.67 +/- 1.42 [%]
Orbital Inclination (inc)	89.81 +/- 2.14
Airmass coefficient 1 (a1)	1.045 +/- 0.05
Airmass coefficient 2 (a2)	-0.016 +/- 0.022
Scatter in the residuals of the lightcurve fit is	4.76 %
Best Comparison Star	#1 - [148, 134]
Optimal Aperture	3.01
Optimal Annulus	13.9
Transit Duration (day)	0.116 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.129 $\pm$ 0.055 vs 0.1457 (deviation 0.3 σ)
Duration fit vs published	0.116 d vs 0.12 d (deviation 0.24 σ)
Mid-time O–C	14.3 min
Depth SNR	2.3
Residual scatter	4.76 %

Light curve

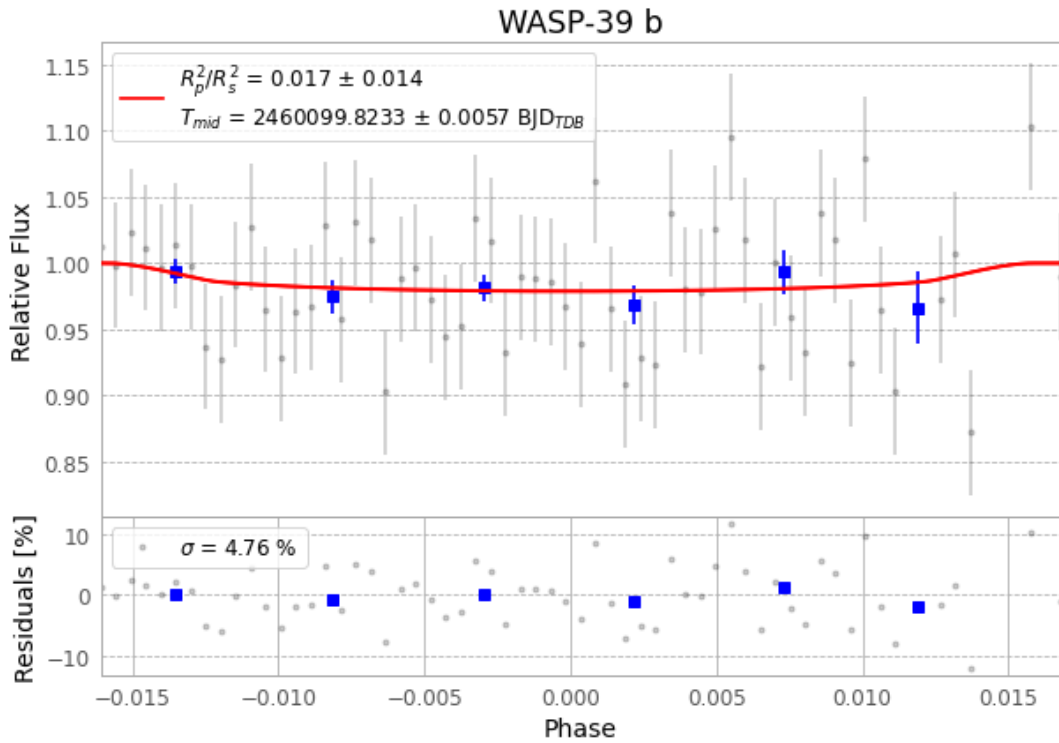


Figure 1 — FinalLightCurve_WASP-39 b_2023-06-03.png (SHA-256 d8f12baa48b39f9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [122, 351], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3e6404c37153494b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-39230604060317.FITS → WASP-39230604094515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-230604.log (a11f4278ca43...), FinalLightCurve_WASP-39 b_2023-06-03.png (d8f12baa48b3...), FinalLightCurve_WASP-39 b_2023-06-03.csv (906dd0dcbec4...)

Compute resources

Wall time 320.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:20Z.